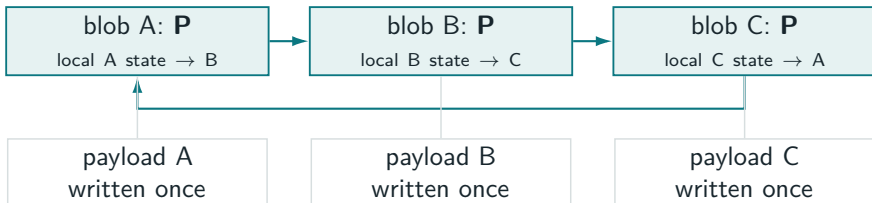


Atomic append-only storage, without writing payload twice

Participant-linked UNO / V2 with caller-delimited payload integrity

Status: opt-in ALPHA API; ordinary Storage and V0/V1 remain unchanged.



concurrent inode barriers closed recovery ring no coordinator file

The contract we actually need

last successfully synchronized epoch



append bytes are visible to this handle

before publication, restart exposes **old**

FAULT MODEL

Any subset of writes not covered by a completed durability barrier may survive.

INDETERMINATE WINDOW

A failed or cancelled sync may recover old or complete new—never a mixture.

PERFORMANCE TARGET

Keep ordinary append behavior: write payload once and pay at the existing sync.

successful sync: complete new epoch



one blob or an all-or-nothing group

after success, restart exposes **new**

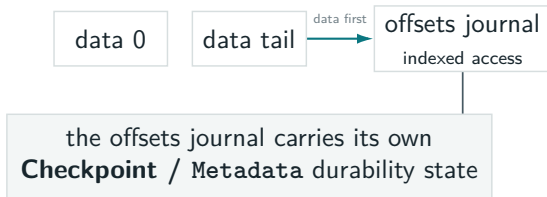
Atomicity is a crash-recovery property; durability remains attached to `sync`.

Before: each journal coordinated its own crash state

LEGACY FIXED JOURNAL



LEGACY VARIABLE JOURNAL

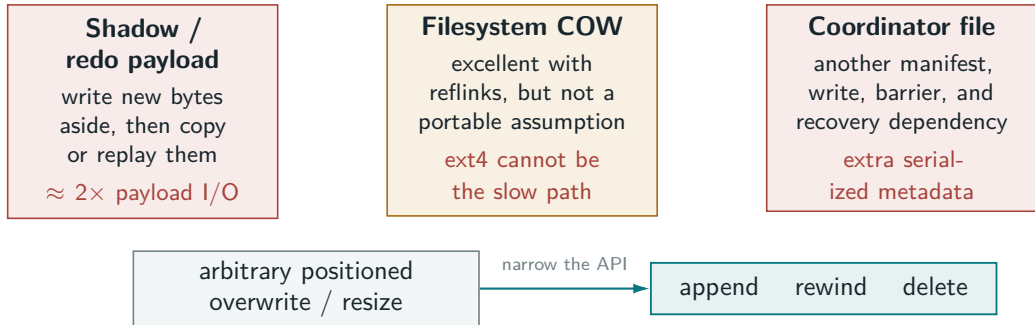


- commit, start_sync, and stronger sync kept data, indexes, and recovery hints in a safe order.
- Main had no filesystem primitive for atomically publishing several blob roots together.

The journals were correct, but every compound structure owned a bespoke durability protocol.

Why generic atomic writes were too expensive

REJECTED DIRECTIONS EXPLORED IN THIS PR—NOT BEHAVIOR SHIPPED ON MAIN



A checksum can detect a torn overwrite; it cannot restore overwritten bytes. Append-only preserves the old prefix, so a root can select a complete visible prefix.

The decisive optimization is an API constraint: preserve old bytes and publish a new length.

V2: one root page, payload at final offsets



R13: 112-byte root header

magic 8 generation 8 encoded length 8
open unit start 8 open CRC32C 4 integrity scheme 4 unit width 4

caller application tag 64 root CRC32C 4

One 2 KiB slot

112 B root + 16 B witness envelope

+ $\leq 1,920$ B local link / unused space

APPEND

Write data and V2-owned footers once at final encoded offsets.

PUBLISH

Alternate root slots; generation parity preserves the immediate fallback.

RECOVER

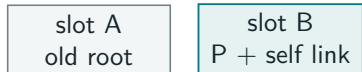
Read bounded roots and, only for an unresolved epoch, checksum a bounded suffix.

No extent map, hole punching, or compaction: a root selects a reachable encoded prefix.

One blob: the durable decision fits in one barrier



covers earlier payload + root + witness
eligible bounded suffix; otherwise payload preflushes first



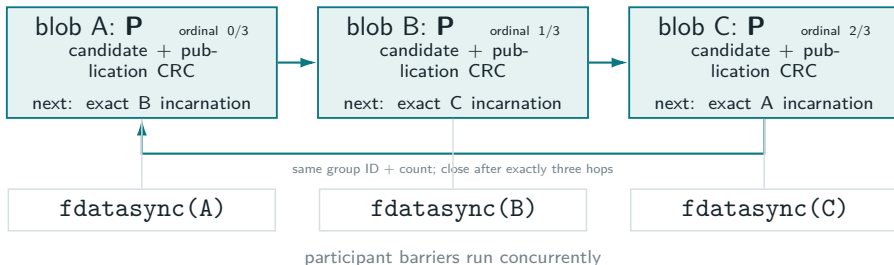
the old root is never overwritten
while publishing its successor

ON RESTART

- If the prepared root, self link, and publication-suffix CRC all validate: expose new.
- Otherwise: expose the preceding committed root.
- No follow-up $P \rightarrow C$ marker and no second durability barrier.

The self-link is the durable decision; later compatible epochs alternate and supersede it.

Several blobs: each root stores one link in a closed ring



WHAT ONE LOCAL WITNESS CARRIES

Fresh 128-bit group ID, count + ordinal, local incarnation/candidate/removal bit, bounded publication-suffix CRC32C, and only the successor's partition/name/incarnation.

Live process: one typed Decision in RAM. On disk: distinct local links only.

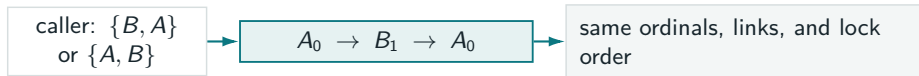
WHAT RESTART PROVES

Opening one unresolved P follows exactly the declared ring, growing state only after each valid link. Only one unique, consecutive, closed ring can publish the complete new group.

No coordinator file and no replicated group descriptor: the participant files are the recovery index.

Overlapping atomic groups

EVERY RING SORTS PARTICIPANT KEYS: (PARTITION, RAW BLOB NAME)



EXACT PARTICIPANT SET REPEATS

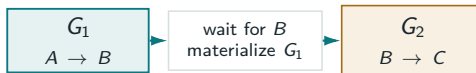


Adjacent generations supersede directly; no $P \rightarrow M$ rewrite.

Protocol policy

disjoint rings need no common order.

PARTICIPANT SET CHANGES



Shared- B serialization is required; eager materialization is current policy.

Prototype policy

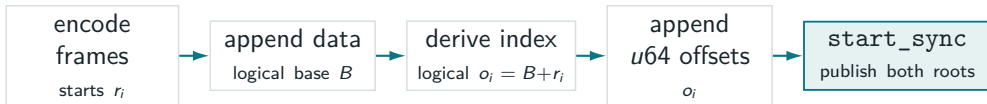
process-local namespace lock
+ singleton carried decision.

Lexicographic links remove caller-order ambiguity and deadlock cycles; they do not create a global commit order.

Dependent writes can be built, then published together

VARIABLE JOURNAL APPEND: SEQUENTIAL STAGING IS ALLOWED

build phase: pending writes may depend on earlier results

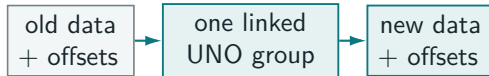


HOW VALUE / IS LOCATED

```
start = offsets[i]
end = offsets[i + 1]
decode data[start..end]
```

Footer-free checked-page coordinates; page geometry maps the root's encoded length to this logical size.

WHAT BECOMES VISIBLE

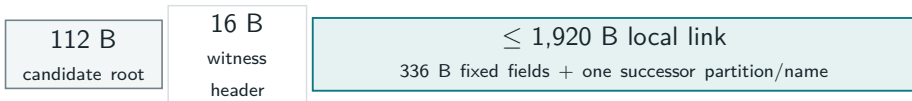


Restart sees the old pair or complete new pair—never mismatched data and offsets lengths.

Stage dependency-aware writes first; one linked UNO group publishes the finished set.

Batch scale: fixed local metadata per participant

EVERY DIRTY PARTICIPANT: ONE 2 KIB SLOT CONTAINS ITS ROOT AND LOCAL WITNESS



PARTICIPANT METADATA



Each participant names only its next exact incarnation. The paths are distributed once around the ring rather than gathered into one record.

PAYLOAD SIZE: NO PROTOCOL BATCH CAP

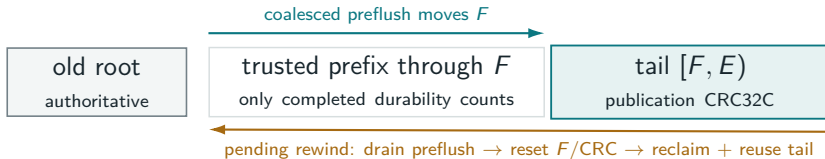


At most **64 MiB aggregate** remains for publication recovery CRC work. Older immutable bytes preflush while appends continue.

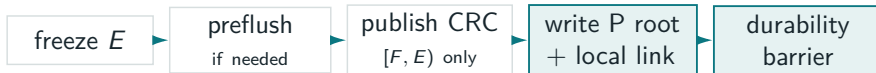
Metadata grows one local link per participant; payload is streamed once rather than copied.

Large epochs bound publication recovery work

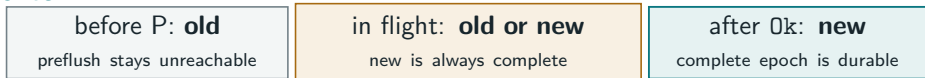
WHILE APPENDING: PAYLOAD IS AT FINAL OFFSETS; NO CANDIDATE ROOT EXISTS YET



AT SYNC: FREEZE ONE EPOCH, THEN PUBLISH



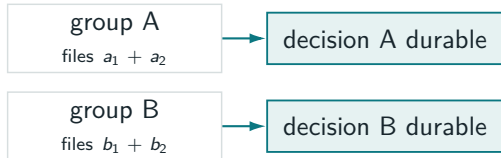
CRASH OUTCOME



Repeat per participant: no payload-byte cap, at most 64 MiB aggregate restart tail, and concurrent final barriers.

Intentional limitation: no global sync order

INDEPENDENT GROUPS



No global order

no root WAL, durable global coordinator, LSN, or commit sequence

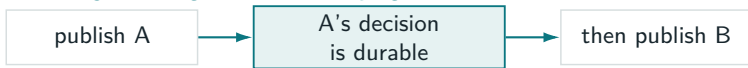
What the protocol avoids

no required persistent global lock
or volume-like filesystem layer

CRASH STATES FOR INDEPENDENTLY STARTED A AND B

old A / old B | new A / old B | **old A / new B is possible** | new A / new B

WHEN B HAS A TEMPORAL DEPENDENCY ON A



old/old, new/old, new/new
never old A / new B

Atomic membership is local. Put a temporal dependency in one group, wait at its edge, or add a higher-level WAL only when a total order is required.

Adoption example: not every file needs V2

FUTURE FREEZER / IMMUTABLE ARCHIVE CONSUMER PATTERN—NOT PART OF THIS PR

KEEP ORDINARY IN-PLACE MATERIALIZATIONS



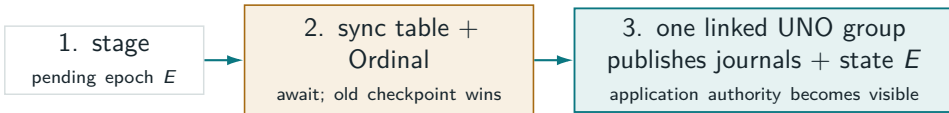
Old checkpoint hides epoch E slots and unset bits.

ADOPT V2 WHERE THE SHAPE IS APPEND-ONLY



One UNO group prevents a mismatched value/index tail.

LOGICAL COMMIT WITH ONE EXPLICIT TEMPORAL DEPENDENCY



before UNO: **old** | in flight: **old or complete new** | after Ok: **new**

If old wins, future materializations remain unreachable; if new wins, their durability was already proven.

Ordinary indexes first; atomic authority second.

Recovery is correct for arbitrary surviving write subsets

What validates after restart	What recovery can prove	Exposed result
No M/T proof; every exact P, local link, incarnation, candidate, and publication suffix validates	One group ID/count; unique consecutive ordinals close after exactly N hops	New roots for the whole group
No M/T proof; any link, incarnation, candidate, or publication suffix is missing/mismatched	The linked group decision is incomplete	Old roots for the whole group
An exact materialized M or tombstone T validates	Installation began only after the P ring was durable	Finish peers; after unlink begins, resume the descending frontier from ordinal 0

TORN ROOTS

Root and witness checksums reject mixed spellings and partial installation. P is invisible to ordinary recovery.

TORN PAYLOAD

A trusted prefix plus publication CRC32C rejects partial visibility with probabilistic checksum assurance.

THREAT BOUNDARY

CRC32C detects crash damage on trusted local disk; it is not tamper authentication.

The protocol never assumes that surviving writes form a prefix in submission order.

Append is the fast path; delete is ordered by phase

APPEND

write final tail once
publish longer root
no relocation or compaction

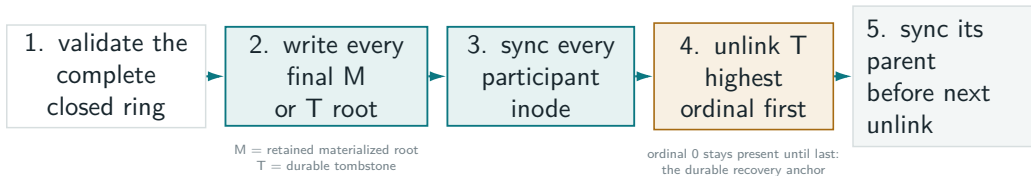
REWIND

publish shorter root
then ftruncate in place
restart repeats a lost truncate

DELETE

durable tombstone
then unlink + directory sync
no separate delete manifest

GROUP DELETE: FINISH THE RING WALK BEFORE CHANGING THE NAMESPACE

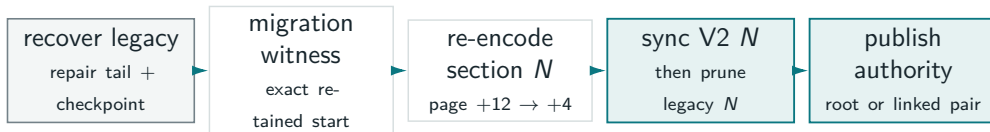


After step 3, each M is independent. Descending, individually durable unlinks leave every remaining T in the forward prefix reachable from ordinal 0.

All final roots first; then a durable descending unlink frontier. Ordinal 0 replaces a delete manifest.

First contiguous V2 open: recover, replace, then switch authority

RESUMABLE, SECTION-AT-A-TIME CONVERSION



CRASH AUTHORITY

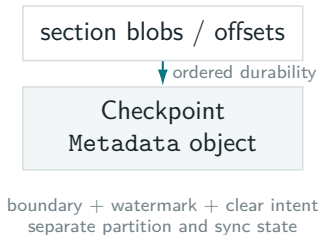


Ordinary V0/V1 opens/formats stay unchanged; generic `migrate_atomic` remains explicit. Peak migration space is V2 plus roughly one overlapping legacy section.

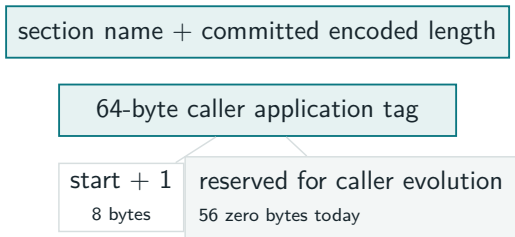
Init returns one V2-only journal; the mixed state exists only inside resumable migration.

Contiguous V2 removes its Metadata checkpoint

BEFORE



NATIVE V2 STEADY STATE

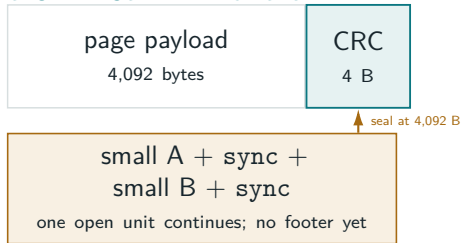


- Variable V2 publishes matching data + offsets roots in the **same UNO group**; neither can advance alone.
- One fixed section or variable pair holds the start; names + encoded lengths map through page geometry to recover logical bounds.
- MutableV2: consuming mutations, `start_sync`, and `sync`—no legacy `commit` split.

Offsets remain as the variable index; the separate checkpoint object disappears.

Integrity units: fixed pages or caller-delimited values

CHUNKED SCHEME: PAGE CACHE

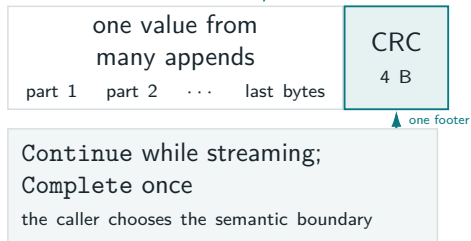


Chunked(4092) binds the fixed payload width. Write calls and syncs do not close a short unit. One footer completes a 4,096-byte physical page, which is then immutable and cacheable.

R13 SELECTED ROOT: EXACT 112-BYTE LAYOUT

magic 8	generation 8	encoded length 8
open unit start 8	open CRC32C 4	integrity scheme 4 unit width 4
caller application tag 64 root CRC32C 4		

VARIABLE SCHEME: GLOB / LONG VALUE

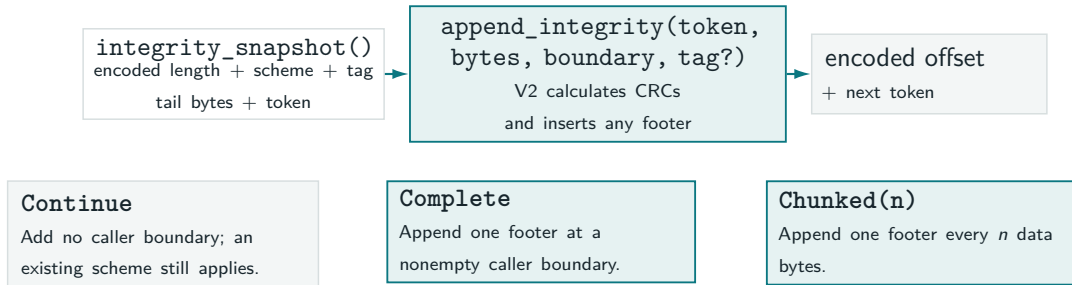


Many appends may form one integrity unit. Complete emits exactly one footer for the open unit. There is no fixed unit width; V2 owns checksum calculation and encoding.

An unfinished integrity unit lives in dedicated root fields—never in the caller's 64-byte application tag.

CRC API: caller boundaries, V2 checksums

ONE COHERENT MUTATION CHAIN; STALE TOKENS FAIL BEFORE PAYLOAD I/O



READ

`read_integrity(unit)` validates data + footer; `snapshot / read_integrity_tail()` validates the open unit. Raw blob reads expose unvalidated encoded bytes.

REWIND

`rewind_integrity(token, end, unit?, tag?)` validates at most the containing unit, then rebuilds the retained prefix's open-unit CRC.

CRC scopes: footer / open-unit field: reads | witness suffix: crash publication | root / link wrapper: metadata

Callers own boundaries and indexes; V2 owns CRCs and stale-writer fencing.

V2 caching: immutable payload pages, mutable RAM

WHAT REWRITES TODAY?

legacy partial physical tip

tail bytes len/CRC A len/CRC B

Append/rewind may rewrite that tip. The legacy tip's two length/CRC records retain the prior logical prefix; this is **not** arbitrary record overwrite.

CURRENT V2 IS DIFFERENT

AtomicSnapshot reads directly
no CacheRef; cache-only reads miss

PLANNED V2 READ PATH

AtomicSnapshot
read

partial tail

retained IoBuf

never enters the
shared page cache

full-page cache
(generation, page)

miss

miss: read page + 4
validate CRC32C

cache logical
page bytes

One cache core still coalesces faults and evicts RAM pages; loaders decode legacy page +12 or atomic page +4.

PROPOSED CACHE INVARIANTS

IDENTITY

One cache ID per opened blob generation; snapshots share it. A recreated name gets a new ID.

APPEND

A completed page is immutable and cacheable. The growing tail stays private and is written only by append.

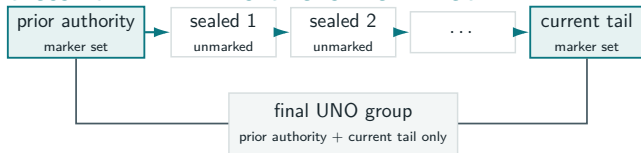
REWIND

Invalidate from the shortened page; publish the shorter root before physical offsets may be reused.

A future cache may replace RAM entries; V2 never overwrites completed payload pages.

Rollover and root recovery stay bounded

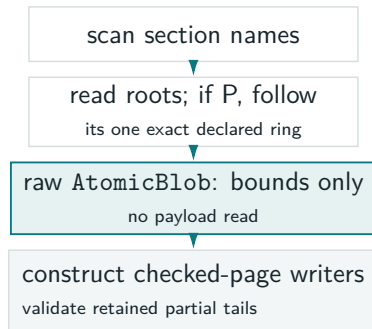
CROSSING MANY FIXED SECTIONS IN ONE EPOCH



Sealed intermediate sections may be durably staged while unreachable; their names fill the interval once the marker publishes.

fixed: **2 roots** variable: **4 roots**
prior/current × data/offsets, all in one group

OPEN / RESTART PATH



Fixed: ≤ 1 tail per retained section. Variable: sealed pairs are complete; active data + offsets: ≤ 2 tails. Full pages remain lazy.

No historical-page scan: raw blob open reads roots; contiguous init reads only retained partial tails.

Raw UNO median: APFS 96.2%; ext4 98.0%

LINKED-WITNESS RAW APPEND VS ORDINARY BLOBS



3.8%

APFS median overhead

2.0%

ext4 median overhead

APFS ratios: .986, .964, .945, .955, .962

ext4 ratios: 1.005, .984, .958, .921, .980

WORKLOAD

- 4 blobs per group
- sixteen contiguous 64 KiB appends per blob
- 4 MiB logical payload per group
- raw append: no integrity-unit footers
- 250 groups per side; 5 paired seeds
- Tokio, 2 workers; inflight 1; sides interleaved

Environments

native macOS APFS; ext4 volume
inside a Linux Docker VM

What remains non-free

local-link construction + locking; one
durability barrier per participant

This isolates group publication; Variable and Chunked(4092) still need paired measurements.

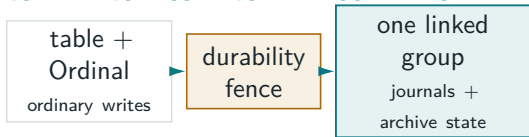
Open question: should every composite store adopt V2?

STOP AT APPEND-ONLY PRIMITIVES



- Existing stores retain Metadata and current durability ordering.
- Smallest compatibility and migration surface.
- Retains serialized checkpoint barriers in composite stores.

CONVERT COMPOSITE COMMIT BOUNDARIES



- Add opt-in V2 commit boundaries to Immutable Archive and peers.
- Fewer serial barriers; payload remains write-once.
- Requires a new format, migration, rollover, and fault tests; linked membership is not root-slot-sized.

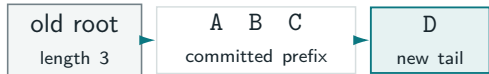
“Convert all” does not mean making every file atomic.

Random-write indexes remain ordinary materializations; V2 owns the append-only journals and the authority that makes those materializations reachable.

Keep V2 targeted—or make it the standard opt-in commit boundary for composite storage?

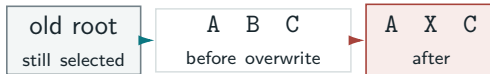
Atomic overwrite is a different storage engine

APPEND-ONLY: THE FALLBACK REMAINS INTACT



The old root still names A B C; selecting it discards D without restoring payload.

OVERWRITE: THE FALLBACK LOSES ITS BYTES



A CRC may reject torn X, but B is gone; the old root cannot restore it.

MAKING THAT OVERWRITE ROLLBACKABLE REQUIRES ANOTHER MECHANISM

COW / shadow pages

Write replacement elsewhere; publish a page map. Adds indirection and reclamation.

Undo / redo log

Persist old bytes or replay data first. Adds payload writes and another ordering protocol.

Volume layer

Version allocation below every store. Adds shared metadata, locking, and recovery work.

Under the fault model, any subset of unsynced overwrite bytes may survive; rollback therefore needs retained old bytes or a separate new copy.

CRC detects but cannot restore overwrites. Composite V2 keeps overwrite-heavy indexes ordinary behind atomic authority.